

Akshat Shukla

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📍 Pune, India

OBJECTIVE

Computer Engineering undergraduate with hands-on experience building end-to-end machine learning and NLP systems using Python, scikit-learn, TensorFlow, PyTorch, and Hugging Face. Experienced in data preprocessing, feature engineering, model evaluation, API deployment, Docker, and CI/CD. Built and evaluated production-oriented ML pipelines with measurable performance improvements and automated testing.

TECHNICAL SKILLS

Languages: Python, Java, SQL, C++, JavaScript

ML/Data: Scikit-learn, Pandas, NumPy, Matplotlib

Deep Learning/NLP: TensorFlow, PyTorch, Hugging Face Transformers

ML Concepts: Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Cross-Validation, NLP, Neural Networks

MLOps/Deployment: FastAPI, Docker, Pytest, GitHub Actions, Streamlit

Tools: Git, GitHub, Linux, Jupyter Notebook

EXPERIENCE

Artificial Intelligence Intern – Codec Technologies

Feb – Mar 2026

- Analyzed financial time-series data using Python and applied preprocessing and feature engineering techniques for ML modeling.
- Trained and evaluated Linear Regression, Ridge Regression, and Random Forest models for stock-price prediction.
- Developed an interactive Streamlit dashboard integrating data collection, preprocessing, visualization, model comparison, and next-day prediction.

PROJECTS

ChurnIQ — Customer Churn Prediction Pipeline [GitHub]

- Built an end-to-end churn prediction pipeline using scikit-learn Pipeline and ColumnTransformer, covering data validation, feature engineering, and preprocessing.
- Selected **Random Forest** (`class_weight='balanced'`) over Logistic Regression and XGBoost via stratified 5-fold CV; achieved **F1: 0.6365** and **ROC-AUC: 0.8435**, with threshold tuned to 0.48 for recall $\geq$ 0.70.
- Deployed the model as a FastAPI service in Docker with Pytest-based automated tests and GitHub Actions CI.

Sentiment Classification — Classical ML vs. Transformer Benchmark [GitHub]

- Benchmarked classical NLP and transformer-based approaches, comparing TF-IDF + Logistic Regression against pretrained DistilBERT for movie-review sentiment classification.
- Achieved **89.68% accuracy** (F1: 0.897) vs. **88.10%** (F1: 0.881) for zero-shot DistilBERT, with the baseline running **~180x faster** (0.69ms vs. 124.9ms) and using **580x less disk** (0.44MB vs. 255MB).
- Performed qualitative error analysis on negation, sarcasm, and nuanced sentiment shifts.

EDUCATION

Sinhgad College of Engineering

B.E. in Computer Engineering

Expected 2027

Pune, India

- **CGPA:** 8.5 / 10

CERTIFICATIONS

- freeCodeCamp — Machine Learning with Python